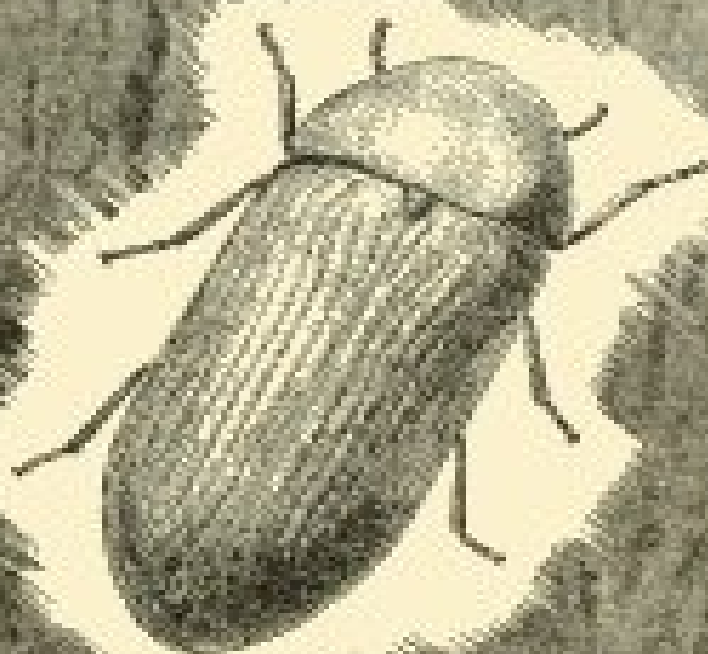


POWDER-POST BEETLES *in Buildings*

what to do about them

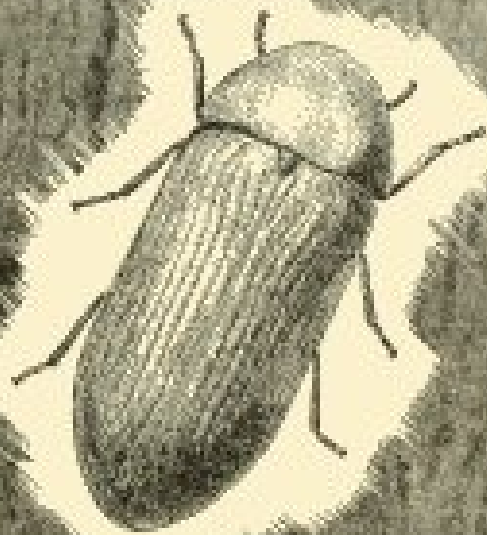


Leaflet No. 358

U. S. DEPARTMENT OF AGRICULTURE

POWDER-POST BEETLES
in Buildings

what to do about them



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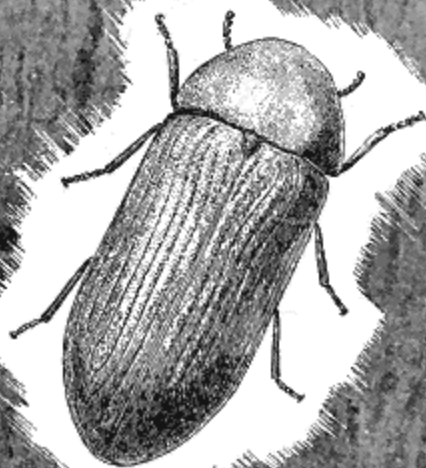
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BEETLES IN BUILDINGS: WHAT TO DO ABOUT THEM ***

POWDER-POST BEETLES
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Leaflet No. 358

U.S. DEPARTMENT OF
AGRICULTURE

Powder-Post Beetles in Buildings

WHAT TO DO ABOUT THEM

Powder-post beetles cause extensive damage to wood in the structure of buildings and to wood products used in homes.

The larvae, or grubs, feed on the starch or the cellulose in wood. As they feed, they bore irregular tunnels through the wood. Borings left in the tunnels have the consistency of powder.

Powder-post beetles can be controlled by applying an insecticide to the infested wood. How it is applied depends on the kind of beetle to be controlled and on the extent and location of the infestation. In some circumstances a spray will suffice; in others, fumigation is necessary.

COMMON AND SCIENTIFIC NAMES OF INSECTS DISCUSSED

Southern lyctus beetle	<i>Lyctus planicollis</i>
Common furniture beetle	<i>Anobium punctatum</i>
Death-watch beetle	<i>Xestobium rufovillosum</i>
Old house borer	<i>Hylotrupes bajulus</i>
Red-shouldered shothole borer	<i>Xylobiops basilaris</i>

Black-horned pine borer	<i>Callidium antennatum</i>
Bamboo powder-post beetle	<i>Dinoderus minutus</i>
Bamboo borer	<i>Chlorophorus annularis</i>

KINDS

Powder-post beetles are divided into five groups—lyctus beetles, anobiid beetles, bostrichid beetles, cerambycid beetles, and cossonid weevils.

Lyctus beetles are reddish to blackish, about $\frac{1}{4}$ inch long, and slightly flattened. One common species is known as the southern lyctus beetle.

Anobiid beetles are similar to lyctus beetles in color and size, but are cylindrical. The common furniture beetle and the death-watch beetle belong in this group.

Bostrichid beetles are reddish, brown, or black; they are $\frac{1}{8}$ to $\frac{3}{4}$ inch long, and cylindrical.

The heads of anobiids and bostrichids are directed downward and are covered by a "hood."

The cerambycids, also known as roundheaded or long-horned beetles, are blue, black, or pale yellow to reddish brown, and often bear distinctive markings on the head or the wing covers. They are $\frac{1}{2}$ to 1 inch long, and vary in shape. The old house borer belongs in this group. It is black, and has grayish markings on the wing covers. It is about $\frac{3}{4}$ inch long.

Cossonid weevils are reddish brown or black, $\frac{1}{8}$ to $\frac{1}{5}$ inch long, and oval to cylindrical. Their heads are drawn out to form a short, broad beak. Cossonids are the only weevils that infest wood; other weevils infest grain or growing crops.

LIFE CYCLE

The adult females lay eggs on or in wood. The eggs hatch into larvae, which develop as they tunnel the wood. The tunneling continues throughout the larval stage, which lasts for months or years, depending on the kind of beetle.

The full-grown larvae transform to pupae, which become adult beetles.

Cover: Adult of an anobiid beetle, *Xyletinus peltatus*, and pine sill damaged by larvae of this beetle.

The adults of many kinds of powder-post beetles bore circular holes to the surface of the wood and emerge; others make oval holes. Some emerge in early spring; others, including the old house borer, emerge in early summer. Mating and egg laying follow.

The life cycle of lyctus beetles varies from 3 months to 1 year, depending on the species and on regional differences in temperature, moisture, and availability of wood of high food value.

The life cycle of anobiid beetles is sometimes completed in about 1 year and sometimes in 2 years.

Most of the bostrichids and cerambycids complete the cycle in about 1 year, but a few of the latter require several years. The old house borer, a cerambycid, requires 5 to 7 years in the North, and less time in the South. In all but a few months of its life cycle, it is in the larval stage. Larvae 3 or 4 years old are often found in the woodwork of buildings only 1 or 2 years old—evidence that stored lumber is a source of infestation.

Cossonid weevils complete the cycle in about 1 year.



Adult of the southern lyctus beetle.

EGG LAYING

The females lay eggs on or in wood that satisfies their egg-laying requirements and the food requirements of the larvae. These requirements differ among the species. Hence different kinds of wood are infested by different species of beetles.

Lyctus beetles infest hardwoods. Cossonid weevils infest softwoods. Some anobiids, bostrichids, and cerambycids infest hardwoods, and some infest softwoods.

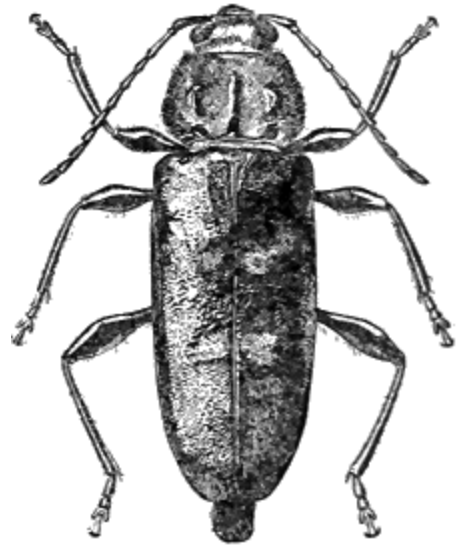
For some species bark is an egg-laying requirement; for others it is not.

Some anobiids infest moist wood containing decay organisms, which appear to aid development of the larvae.

The porosity of the wood, the moisture content, the starch or cellulose content—these also relate to egg laying.

The adults of some species remain on the wood from which they emerge, and reinfest it. Others crawl or fly to similar wood and attack it.

A beetle may lay only a few eggs or as many as 60 or 70; the number depends on the species of beetle and on the condition of the wood.



Adult of the old house borer.



Adult of the red-shouldered shothole borer, a bostrichid beetle.

Infestations in Hardwoods

SEASONED HARDWOODS.—Beetles that infest seasoned hardwoods lay eggs on or in the wood from which they emerge. Hence the same wood may be infested repeatedly. These beetles do not require bark for egg laying.



Oak board damaged by larvae of the southern lyctus beetle.

Lyctus beetles lay eggs in exposed pores in the sapwood of oak, ash, hickory, walnut, pecan, gum, persimmon, maple, cherry, and a few other hardwood species. The larvae are found in hardwood flooring, furniture, picture frames, and trim.

Anobiid and cerambycid beetles lay eggs in crevices on the surface of the wood; cerambycids also lay eggs in emergence holes.

Bostrichid beetles that bore into seasoned hardwoods often bore chambers at the sides of their tunnels and lay eggs in them.

UNSEASONED HARDWOODS.—Bostrichids that infest green and seasoning hardwoods lay eggs in pores leading from the entrance tunnels. The red-shouldered shothole borer is one of these. Bostrichids frequently attack bark-covered wood. They usually do not reinfest the wood from which they emerge.

Infestations in Softwoods

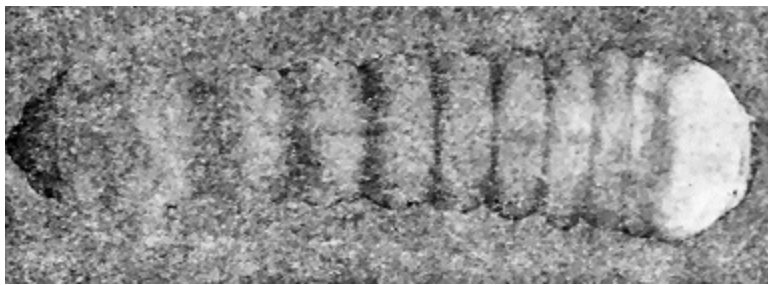
SEASONED SOFTWOODS.—Some kinds of beetles infest seasoned softwood lumber used in the structure of buildings.

Certain of the bostrichids require bark for egg laying. They infest lumber that has bark on the edges. Such lumber is often used in framing, subflooring, and roofing. The females lay eggs in the bark. The larvae of some species complete their development in it; others go to the adjacent wood to do so. The adult females may reinfest the wood from which they emerge.

Other beetles that infest seasoned softwoods do not require bark for egg laying. Among them are the old house borer and the cossonid weevils. The females lay eggs in crevices in the wood. They may infest the same wood repeatedly.

The old house borer infests framing and subflooring. It is found in old and new construction and in stored lumber. Cossonid weevils are found in pine flooring, baseboards, wall stops, and roof trusses.

UNSEASONED SOFTWOODS.—The beetles that infest green and seasoning softwoods include several cerambycids—for example, the black-horned pine borer—and a few bostrichids. Most of these beetles require bark for egg laying. The adult females usually do not reinfest the wood from which they emerge; they seek other unseasoned wood and lay eggs in the bark.



Larva of the old house borer. Enlarged about 5×.

Infestations in Bamboo

Certain species of bostrichid, cerambycid, and lyctus beetles infest bamboo products such as furniture. Venetian blinds, draw drapes, baskets, fans, and fishing poles.

Plant-quarantine inspectors at United States ports of entry often intercept the beetles in bamboo products imported from the Far East. Among the beetles intercepted are the bamboo powder-post beetle, a bostrichid, and the bamboo borer, a cerambycid. These beetles and the lyctus beetles may reinfest the material from which they emerge.



Pine floor joist damaged by larvae of the old house borer

DAMAGE

The increase in building activity that followed World War II brought the destructiveness of powder-post beetles to widespread attention. The more buildings, the more damage.

The old house borer is rapidly spreading over the eastern half of the United States, and is found more and more in lumber used in new construction. It is becoming one of the most serious pests of wood in buildings.

Powder-post beetles damage wood by tunneling and by cutting surface holes.

Tunneling is done by the larvae of all kinds of powder-post beetles, and by adult bostrichids and cossonid weevils. The larvae of anobiid beetles leave elongate pellets of excreted wood in their tunnels. Other larvae leave powderlike borings. Severely damaged wood becomes a crumbly mass of pellets or of borings, and sometimes has a honeycombed appearance.

Since the insects do most of their tunneling in sapwood, structural damage is rare in large timbers.

Surface holes mar the appearance of wood. Those most commonly seen are made by adult beetles when they emerge from the wood. Those made by many of the smaller beetles are about $\frac{1}{8}$ inch in diameter. They give wood the appearance of having been hit by buckshot, and are called shotholes. Larger beetles make holes $\frac{1}{4}$ to $\frac{3}{8}$ inch in diameter.

Emergence holes made by the old house borer are oval. The lengthwise measurement is about $\frac{1}{4}$ inch.

The larvae of some species, including the lyctus beetles, make holes to the surface of the wood and expel borings through them. The larvae of the old house borer seldom pierce the surface; they pack their borings behind them.

DETECTION

Aside from observing tunneled wood, there are several ways of detecting powder-post beetle damage.

You may see surface holes, or borings that have been ejected from them. The borings may be in piles near the holes or, where timbers overhead are attacked, they may be on the floor below.

You may see adult beetles crawling on the wood.

In quiet surroundings you may hear a rasping or ticking sound that is made by tunneling larvae. This is the best way to detect larvae of the old house borer. Their ticking is sometimes audible several feet from where they are working. By determining the source of the sound and by probing the wood with a sharp tool, you can locate the tunnels. A further aid is to look for blistered wood.

Sometimes the larvae of cerambycid beetles can be heard working beneath bark on green or partially seasoned wood. They frequently are found in firewood. They resemble the old house borer, but they have different habits and do not infest seasoned wood. When the adults emerge, they seek more green wood to attack. If they cannot escape from the building, they die in a few days.

WHAT TO DO

Consider whether the control problem is one that you can handle yourself or whether it is one that requires the services of a pest-control operator.

If you have a severe infestation, or one that is behind paneled or plastered walls, or in some other hard-to-reach place, it may be that only an experienced pest-control operator with the proper equipment can do a satisfactory job.

If the infestation is light and easy to reach, you may be able to handle the control job yourself. Spraying or brush-coating the wood with an insecticide may suffice.

If wood has been so badly damaged that its structural strength is impaired, it should be replaced. Coordinate the repair work with the control work.

Spraying or Brushing

Beetles that make numerous small holes in the surface of wood are controlled by saturating the wood with a deodorized kerosene, such as deobase or ultrasene, which contains either 5 percent DDT, 2 percent chlordane, 0.5 percent dieldrin, or 0.5 percent lindane.

With 1 gallon of any of these materials you can treat at least 100 square feet of wood surface. Use no more of the material in a single application than is needed to wet the surface. Repeat the application until the wood is saturated. Apply with a garden sprayer or a paintbrush.

The treatment is also effective against infestations in furniture and other wood products.

For best results in treating any kind of material, first remove the finish. This is especially desirable when you treat a large area, such as a floor, but it may not be desirable when you treat ornamental products. Where an infestation in a floor is so small that removal of the finish is undesirable, go

ahead and treat the floor, but do not walk on it until it has dried. The drying takes several hours.

The old house borer and other beetles that do not make entrance holes in the surface of the wood can also be controlled by surface applications of these chemicals. Larger quantities must be used, however, especially where thicker wood is concerned, such as floor joists, studs, and roof rafters. For such materials, at least 1.5 pints of the insecticide per cubic foot of wood is needed to reach the beetle larvae. This is equivalent to treating the exposed surfaces of 7 linear feet of a 2- by 10-inch floor joist, 12 feet of a 2- by 6-inch roof rafter, 18 feet of 2- by 4-inch studding, or 12 square feet of subflooring. Even with this thorough treatment, it may take from 3 weeks to 3 months before all the larvae are reached and killed.

The homeowner can best apply such a quantity of insecticide by using a sprayer or a pint-size plastic wash bottle of the squeeze type, equipped with a tube that will produce a pinpoint stream. This makes it possible to apply small enough quantities to cover the surface without causing the liquid to run off. Repeated applications at different times are necessary for complete absorption by the wood and effective penetration. A pest-control operator can obtain results quickly by injecting the chemical under pressure through boreholes spaced at short intervals in the infested wood.

Slow-Diffusion Method

Where pressure equipment is not available and you want to use the bore-hole or slow-diffusion method on isolated infestations in wood that is readily accessible, you can use the following procedure:

1. Bore a downslanting hole about $\frac{1}{2}$ inch in diameter and 1 inch deep into the wood near the place where a larva is working.
2. Insert a tight-fitting metal or glass tube in the hole. Let about $\frac{1}{4}$ inch of the tube project.
3. Place one end of a rubber tube over the projecting portion of the metal or glass tube, and connect the other end with a funnel. The funnel should be supported so that it will stay in place until the treatment has been completed.
4. Pour insecticide into the funnel. Use any of the preparations mentioned in the discussion of spraying and brushing. Leave the equipment in place about 48 hours, or until

the insecticide is taken up. The insecticide flows into the infested wood, is absorbed for a distance of several inches around the hole, and reaches and kills the larva.

If several larvae are working in places as much as 18 inches apart, individual treatment is needed.

Fumigation

Treatment of a heavy infestation of powder-post beetles by spraying or brushing or by the slow-diffusion method is impractical where the wood is hard to reach; particularly when it is located behind plastered walls or covered by paneling or insulation.

Such an infestation can be controlled by fumigation—but *this work must be done by a licensed fumigator.*

Hydrocyanic acid gas or methyl bromide gas is used in fumigating. Each is very poisonous, and heavy dosages have to be used.

After fumigation, the building must be aired to make it safe for reoccupancy. The fumigating and the airing require 2 to 4 days.

Fumigation methods of the "home-remedy" type—such as burning sulfur candles—are ineffective because the fumigant does not penetrate the infested wood.

PRECAUTIONS

The insecticides mentioned in this leaflet are poisonous to people and to animals. Keep them where children and pets cannot reach them. Do not let them come in contact with the body. When applying them, wear rubber gloves, a cellulose acetate type of face-guard, and a rubber or leather apron. When spraying an overhead area, do not stand directly under it.

GUARDING AGAINST EXPLOSIONS.—Apply a coarse spray. It is effective and safe to use. A concentrated fine mist is ineffective and may explode in the presence of a spark or flame. Do not smoke while spraying. Shut off pilot lights.

PRESSURE.—In spraying, use only enough pressure to wet the surface of the wood. Too much pressure may cause spray to splash onto the body. If you apply the insecticide in an enclosed place, make sure that the air can circulate freely. Keep doors and windows open; if necessary, use a fan. The purpose is to prevent odors from accumulating in the building.

If you treat a floor that has a finished ceiling under it—for example, the floor of a second-story room—avoid applying too much material. If you do, it may run through to the ceiling and cause a stain. Removal of the floor finish before applying the insecticide makes it unnecessary to apply a large amount; the wood absorbs most of it.

If you apply the oil solution to parquet flooring, do it very lightly. If you apply too much, the oil may dissolve the asphalt in which the flooring is set. The dissolved asphalt may stain the floor. Around the edges of the floor, it may "creep up" and damage the walls.

If you have applied insecticide with a garden sprayer, rinse out the tank and hose before putting the sprayer away. This prevents the gaskets from shriveling and keeps oil from getting on plants the next time the sprayer is used in the garden.

FOOD CABINETS.—In treating a food cabinet, do not use an insecticide because it might contaminate food. Use deodorized kerosene alone. It has little odor and evaporates quickly. Take the cabinet outdoors to treat it, and leave it there until the kerosene odor disappears.

FUMIGATION.—The fact that fumigation work must be done by a licensed operator is stated on [page 7](#).

Only isolated buildings can be safely fumigated. If an apartment, a row house, or part of a duplex were fumigated, the gas might penetrate walls and kill persons in an adjoining unit.

Methyl bromide gas is odorless, but in the presence of articles having a high sulfur content, or containing animal matter, sponge rubber, or iodized salt, a chemical change takes place and a garliclike odor is apt to be given off later. Before this gas is used, see whether such articles are in the building; if they are, remove them. This need not be done if hydrocyanic

acid gas is used. However, hydrocyanic acid gas tarnishes silver and spoils unexposed photographic film; before it is used, remove silver and film.

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